

Eliana Ostrovsky

AI Engineering Student

Machine Learning • Computer Vision • Reinforcement Learning • LLM Systems

Buenos Aires, Argentina

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Professional Summary

AI Engineering student with hands-on experience developing and deploying machine learning systems, LLM-based workflows, and autonomous agents. Experienced in building production-ready AI tools, integrating cloud infrastructure, and translating business requirements into technical implementations. Ranked in the Top 9% of the AI Engineering program at Universidad de San Andrés.

Professional Experience

AI Engineer | Bouldertech.ia Oct 2025 – Present

- Design and implement custom AI solutions using modern ML and LLM technologies.
- Deploy and benchmark open-source LLMs (Qwen family) and vision models using Ollama on dedicated servers.
- Build RAG pipelines and autonomous web navigation agents using OpenCrawl for automated data extraction.
- Containerize AI systems using Docker to ensure reproducible environments.

Data Analyst & AI Developer | Alephee May 2025 – Present

- Lead the development of internal AI tools and automation workflows.
- Build LLM-powered orchestration agents in n8n integrated with Microsoft Teams, SharePoint, and Azure.
- Implement workflows generating structured JSON outputs with integrated validation and security layers.
- Query and manage large datasets using Microsoft SQL Server.
- Prototype cloud solutions using AWS (Bedrock, SageMaker, S3, EC2).

ML Intern | DHC 2024

- Built predictive models using Machine Learning and Feature Engineering to optimize operations in hospitals and medical centers.

Teaching Assistant | Universidad de San Andrés 2024 – 2025

- Physics I and Calculus III (Análisis Matemático III). Prepared and delivered classes, graded exams and lab reports.

Engineering Projects

Trademark Opposition Management Platform

- Building a full-stack web platform for Argentine lawyers that automates weekly INPI (Instituto Nacional de la Propiedad Industrial) bulletin processing and trademark conflict detection.
- Compare new trademark filings against client portfolios using multiple complementary algorithms: fuzzy string matching, phonetic similarity, semantic embeddings, CNN-based visual similarity, and LLM reasoning.
- Develop a web interface for lawyers to review flagged conflicts, dispatch notification emails to clients, and track client responses throughout the opposition workflow.

LLM Interpretability and Behavior in Multi-Agent Social Simulation

- Team research project studying how open-source LLMs (Llama-3.3-70B, Gemma-3-27B, Qwen3-8B) reason and influence each other inside multi-agent social simulations; built the per-agent model routing and per-call telemetry that made the comparison possible.
- Characterized architecture-specific failure modes in open-ended debate — epistemic echo chambers, factual impermeability where pre-training priors override contrary evidence, and evasive neutral consensus — traced to pre-training coverage and alignment.

- Quantified generative behavior independently of task accuracy (Self-BLEU, distinct- n , Vendi Score, entropy) and intra-persona ideological drift via parametric interviews at simulation checkpoints; found heterogeneous populations collapse into semantic deadlock without exogenous information.

Lane Detection via Teacher–Student Distillation under Domain Shift

- Accepted at the Jornadas Argentinas de Robótica (JAR 2026).
- Designed a teacher–student domain adaptation scheme for lane detection on the PAMPA urban dataset using pseudo-labeling, without retraining the teacher’s weights.
- Implemented two student variants based on CLRRerNet: image-only and a temporal extension with ConvLSTM for inter-frame coherence.
- Quantitatively characterized the specialization vs. forgetting trade-off: urban F1 of 34.9% vs. highway accuracy drop from 96.4% to 18.2%.

Car Price Prediction

- Performed extensive data engineering including cleaning, feature extraction, and transformation of raw automotive datasets.
- Trained and fine-tuned an XGBoost model with systematic hyperparameter optimization to maximize prediction accuracy.

Autonomous Navigation (PPO)

- Implemented a Proximal Policy Optimization model for self-driving agents in a Unity simulation environment.

Research

Research Assistant Program Participant – UdeSA Autonomous Vehicle Project | UdeSA Argentina’s first research autonomous street vehicle (udesarobotics.github.io). Worked with the PAMPA dataset, the first autonomous driving dataset in the region (CONICET).

Anti-inflammatory Effect of Nanoparticles on Blood Vessel Endothelium

- Research internship at the Experimental Thrombosis Laboratory, Institute of Experimental Medicine (CONICET – Academia Nacional de Medicina, Buenos Aires).
- Investigators: Charó Nancy, Schattner Mirta.

Education

B.S. in AI Engineering | Universidad de San Andrés

- Expected Graduation: July 2027.
- Top 9% of the AI Engineering program.
- Top 10% university-wide.
- GPA: 8.72 / 10.0.

High School Diploma in Natural Sciences (Chemistry & Biotech) | ORT Argentina

- Graduated with Honors.

Technical Skills

- **Programming:** Python, SQL, C++, Java, C, JavaScript, Haskell.
- **Machine Learning:** PyTorch, Computer Vision, Reinforcement Learning (PPO), LLM Systems, RAG, LLM Evaluation.
- **Infrastructure:** Docker, Git, n8n, ROS2, Gazebo, Unity, Ollama, Neo4j.
- **Cloud:** AWS (Bedrock, SageMaker, S3, EC2), Microsoft 365 (SharePoint, Teams, Azure), Google Cloud.
- **Languages:** Spanish (Native), English (C2), Hebrew (Basic).

Additional Experience

- **Cadena** – Active participant in this international NGO.
- **Israel Challenge, MASA** – 4.5-month cultural immersion program in Israel.
- **BBYO Exchange** – 2-week international relations program in Denver, Colorado, USA.
- **ORT Study Trip** – 2-week English language and culture program in London, UK.